

Weather Alerts (Post Mid eval)

Model Exploration & Benchmarking:

I did comparative analysis and benchmarking of three machine learning architectures designed for multivariate, multi-sector time-series weather forecasting. The objective was to evaluate the baseline Long Short-Term Memory (LSTM) network against a Temporal Fusion Transformer (TFT) and an Extreme Gradient Boosting (XGBoost) model. The models were evaluated on their ability to predict critical meteorological hazards across the national airspace over a 12-hour tactical horizon.

The predictive engine processes four specific meteorological variables critical to aviation safety and dynamic flight routing.

- **Temperature (°C):** Defines the core thermal trend and is required for identifying extreme heat hazards that affect air density and aircraft lift performance.
- **Relative Humidity (%):** A primary indicator of atmospheric moisture, utilized to anticipate dew point saturation and low-visibility cloud cover.
- **Wind Speed (m/s):** The most critical flight safety metric, used directly in the routing algorithm to trigger high-wind threshold alerts and sector evacuations.
- **Precipitation (mm):** Evaluated to identify heavy rainfall and severe storm cells along designated flight corridors.

Horizon Configuration: The models operate on a sliding window architecture. An input lag of 48 hours (capturing two complete diurnal cycles) establishes the baseline momentum, outputting a 12-hour multi-step forecast window required for zero-latency tactical route planning.

To ensure a chronological train/test split was executed on a 5-year historical dataset (2021–2025) spanning five major provincial capitals.

- **Training Set:** 4 years and 11 months (43,104 hours).
- **Validation Set (Blind Test):** The final 30 days (720 hours) were withheld from the models to prevent data leakage and evaluate real-world inference capability.
- **Evaluation Metric:** Mean Absolute Error (MAE) was calculated after inverse-transforming the scaled outputs back to real-world meteorological units.

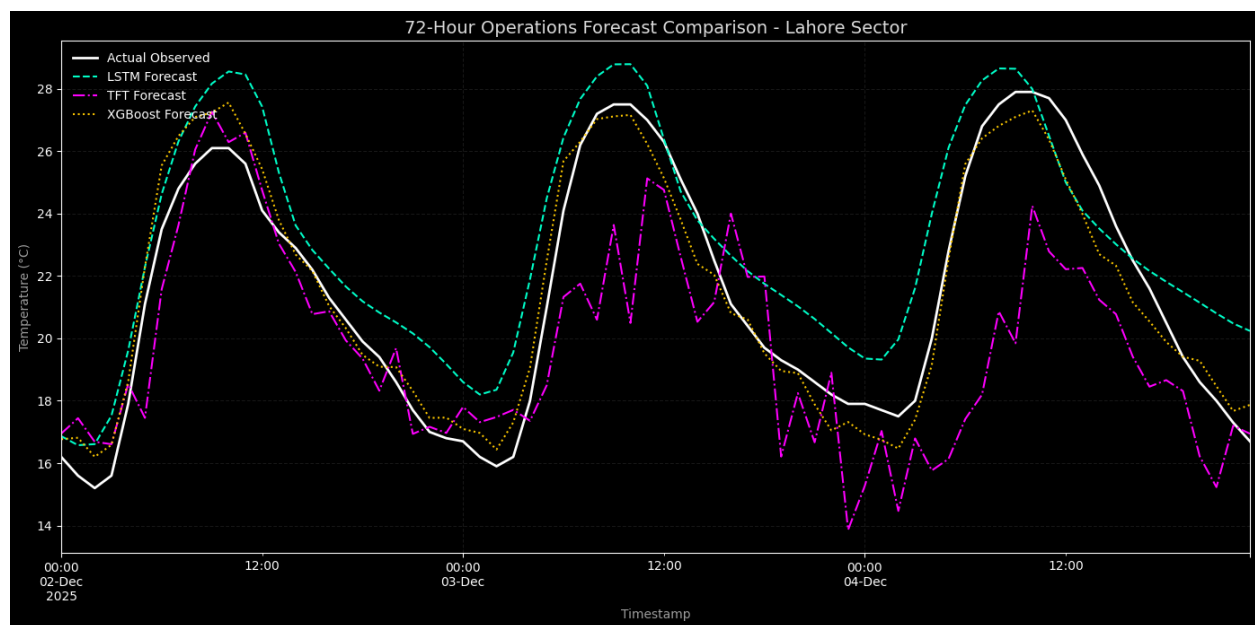
The benchmarking yielded the following Average MAE scores across the 720-hour test horizon:

- **LSTM (Baseline):** 10.13
- **Temporal Fusion Transformer (TFT):** 7.02
- **XGBoost:** 5.78

Conclusion & Selection:

The XGBoost model was selected as the final production architecture. While the baseline LSTM successfully preserved the diurnal wave shape, it suffered from a systematic offset error. The TFT, exhibited high variance and severely underpredicted daytime thermal peaks due to the lack of encoded future covariates.

XGBoost achieved the highest accuracy (MAE: 5.78) because its tabular decision boundaries efficiently map historical feature interactions directly to target thresholds. It explicitly captured peak amplitudes and trough depths without the phase delays common in recurrent networks, making it the most reliable engine for tactical airspace routing.



High-Frequency Active Radar (Nowcasting)

- Implemented a short-burst micro-forecasting engine to complement the 12-hour macro-forecasting pipeline, mimicking modern tactical aviation radar systems.
- **Temporal Resolution:** Transitioned from 1-hour intervals to high-frequency 15-minute telemetry steps (minutely_15).
- **Lookback & Horizon Architecture:**
 - **Lookback Window (t - 4 hours):** Ingests 16 steps of 15-minute historical telemetry to capture immediate storm-front velocity and atmospheric pressure shifts.
 - **Prediction Horizon (t + 60minutes):** Predicts 4 steps ahead (30–60 minutes) to provide real-time tactical hazard alerts on the active radar UI.
- **Model Selection:** Trained a lightweight, GPU-accelerated XGBoost model (tree_method="hist") on 90 days of high-frequency multi-sector weather data.



Engine	Resolution	Lookback	Prediction Horizon	Key Features
Macro Engine (Thread 1)	1-Hour	48 Hours	Next 12 Hours	Raw Hourly Telemetry
Active Radar (Thread 2)	15-Min	4 Hours	Next 60 Mins	Telemetry + Derivatives + Sin/Cos Encodings